1. Retrieve the names (fname,lname) of employees in department 3 who work more than 22 hours per week on the 'AGI' project. (“AGI” is project name.)

   AGI_Project ← σPNAME = “AGI” (PROJECT)

   AGI_Proj_No ← AGI_Project Join on (PNUMBER=PNO) WORKS_ON

   Hours_worked ← σHOURS > 22 (AGI_Proj_No)

   22_hours ← EMPLOYEE Join on SSN=ESSN Hours_worked

   Dept3_Emps ← σDNO=3 (22_hours)

   Result ← πFNAME,LNAME (Dept3_Emps)

2. List the name (dependent_name) of female dependents, if the dependents depend on the employees who work for the ‘Marketing’ department. (“Marketing” is department name).

   Marketing_Dept ←σDNAME = ”Marketing”(DEPARTMENT)

   Marketing_Emps ← Marketing_Dept join on DNUMBER = DNO EMPLOYEE

   Marketing_Emp_Dependents ← Marketing_Emps join on SSN=ESSN DEPENDENT

   Female_Dependents ← σ SEX=’F’(Marketing_Emp_Dependents)

   Result ← πDEPENDENT_NAME(Female_Dependents)

3. Find the names (fname,lname) of employees that are directly supervised by 'Annie Smith'.

   AnnieSmith←σFNAME=*'Annie'*∧LNAME=*'Smith'*(EMPLOYEE)

   SupervisedEmployees←*AnnieSmith* SSN=SUPERSSN (Employee))

   Result←πFNAME, LNAME(SupervisedEmployees)

4. Retrieve the names (fname,lname) of all employees who do not work on any project.

   EmployeesOnProjects←πESSN(WORKS_ON)

   AllEmployees←πSSN(EMPLOYEE)

   EmployeesNotOnProjects←(AllEmployees−EmployeesOnProjects)

   Result←π FNAME, LNAME